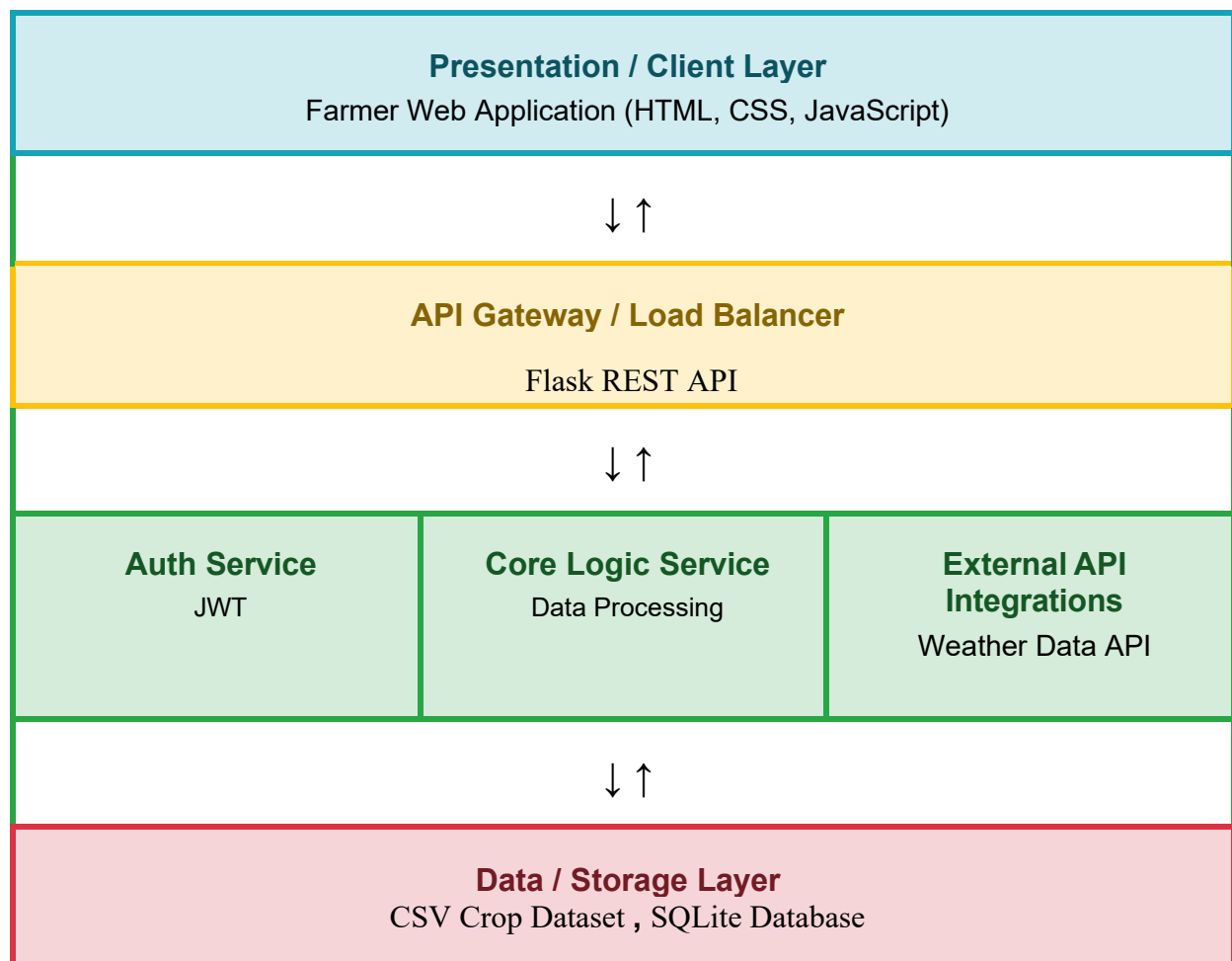


Solution Architecture

Date	30 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User interface for farmers and administrators to enter soil and environmental data and view recommendations.	HTML, CSS, JavaScript
API Gateway	Receives requests from the client and communicates with backend services.	Flask REST API, Python
Microservices	Performs crop prediction, soil analysis, environmental processing, and recommendation generation.	Python, Scikit-learn, Pandas, NumPy
Database	Stores crop dataset, prediction results, and application records.	SQLite, CSV Dataset